Project Report

Project Title: Urban Development & Public Sentiment Analysis

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1. Project Overview

This Power BI project focuses on analyzing urban infrastructure, citizen feedback, and public service efficiency using synthetic city data. The dashboard integrates information from multiple sources such as civic service requests, social media sentiment, and public transit events to uncover relationships between citizen satisfaction, urban maintenance, and response time.

2. Project Objectives

• To visualize and analyze the distribution of civic service requests by category and neighborhood.
• To assess public sentiment from social media posts related to infrastructure and city services.
• To evaluate public transit performance using average delay data.
• To develop a “NeedScore” that ranks neighborhoods based on issue frequency, response time, and sentiment.
• To provide an interactive dashboard for data-driven decision-making in urban planning.

3. Dataset Description

The dataset (urban_public_sentiment_dataset.xlsx) contains six related tables representing simulated city operations data for one week.

4. Data Cleaning (Using Power Query)

Data preparation was performed using Power Query. Transformations included removing duplicates, converting date columns, handling nulls, and calculating resolution time. Cleaned datasets were used to generate consistent visuals.

5. Data Modeling

A star schema model was created in Power BI with neighborhoods as the central dimension table. Relationships were built on neighborhood_id. DAX measures were used to calculate KPIs such as Total Requests, Avg Resolution Hours, Negative Sent %, and NeedScore.

6. Dashboard and Visualization Summary

The dashboard consists of four pages:

Page 1 — Service Requests Overview: KPIs, bar chart by category, and Azure heatmap.
Page 2 — Public Sentiment: Sentiment trends and map visualization.

Page 3 — Neighborhood NeedScore Dashboard: Ranked neighborhoods and NeedScore visualization.
Page 4 — Transit Events & Delays: Delay KPIs, trends, and map of transit performance.

7. Coding Part (Dataset Generation)

Python was used to generate the synthetic dataset using Pandas, NumPy, and XlsxWriter. It created data for service requests, social posts with sentiment scores, and transit delays. The final Excel dataset contained six sheets for Power BI analysis.

8. Outcome

A fully functional Power BI dashboard integrating service request data, public sentiment, and transit analytics was created. It enables data-driven decisions, highlighting high-need neighborhoods and supporting city planning through real-time insights.